



Bahir Dar University
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Advanced Applied Mathematics III (Math 2042) worksheet II

1. Find the arc length, the unit tangent vector, the unit normal and the curvature of a curve on the given interval whose parameterization is given below
 - a) $r(t) = (1 - 2\cos t)i + 2\sin t j - 12tk$; $0 \leq t \leq 1$
 - b) $r(t) = \cos^3 t i + \sin^3 t j + 3k$; $0 \leq t \leq \frac{\pi}{2}$
2. Determine the curl and divergence of $F(x, y, z) = e^{-xy}i + xe^{-yz}j + xe^{-xz}k$
3. Determine whether F conservative or not and find the potential function if F conservative.
 - a) $F(x, y) = ye^{xy}i + (xe^{xy} + \cos y)j$ c) $F(x, y, z) = (2x + 3)i + (2y + z)j + yk$
 - b) $F(x, y, z) = 4xe^z i + \cos y j + 2x^2 e^z k$
4. Evaluate the following line integrals
 - a) $\int_C (y + 2xe^y) dx + (x + x^2 e^y) dy$, where $C: r(t) = \sqrt{t}i + \ln t j + tk$, for $1 \leq t \leq 4$
 - b) $\int_C (x + y) ds$, where $C: r(t) = (e^t + 1)i + (e^t - 1)j$, for $0 \leq t \leq \ln 2$
 - c) $\int_C (2xy - 5yz) ds$ where C is the line segment from $(0, 1, 0)$ to $(0, 3, 2)$.
 - d) $\int_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$ where C is a circle of radius 2
5. Evaluate $\int_C F \cdot dr$ where;
 - a) $F(x, y, z) = zi - yj - xk$ and $C: r(t) = 5i - \sin t j - \cos t k$, for $0 \leq t \leq \frac{\pi}{4}$
 - b) $F(x, y, z) = ye^{x^2} i + x \cos y^3 e^{y^2} k$ and $C: r(t) = e^t i + e^t j + 3k$, for $0 \leq t \leq 1$
 - c) $F(x, y, z) = \frac{8x^3}{1+x^4+y^4+z^4} i + \frac{8y^3}{1+x^4+y^4+z^4} j + \frac{8z^3}{1+x^4+y^4+z^4} k$ and $C: r(t) = t^2 i + t^4 j + t^6 k$, for $0 \leq t \leq 2$.
6. Use Green's theorem to evaluate (C is oriented positively)
 - a) $\int_C 2x^3 y dx - 3x^2 y^2 dy$; where C is a triangle with vertices $(0, 1)$, $(3, 4)$ & $(2, 7)$
 - b) $\oint_C y \sin x dx - \cos x dy$; where C is the path from $(3, 0)$ to $(-3, 0)$ along the curve $y = \sqrt{9 - x^2}$ and from $(-3, 0)$ to $(3, 0)$ along the x -axis.
 - c) $\oint_C (y - x) dx + (2x - y) dy$; where C is the boundary of the region between the graphs of $y = x$ and $y = x^2 - 2x$.
 - d) $\oint_C e^{e^x} - 2y dx + \tan^{-1}(ye^{-y} + 1) dy$, where C is the ellipse $9x^2 + 4y^2 = 36$.
7. Evaluate the following surface integrals
 - a) $\iint_S \sqrt{4x^2 + 4y^2 + 1} dS$, where S is part of the paraboloid $z = x^2 + y^2$ below the plane $y = z$
 - b) $\iint_S 5zy^2 dS$, where S is the portion of the cone $z^2 = 2x^2 + 2y^2$ that lies between the planes $z = 1$ and $z = 2$.

- c) $\iint_S z(x^2 + y^2) dS$, where S is the hemisphere $\sqrt{x^2 + y^2 + z^2} = 2$.
- d) $\iint_S \sqrt{1 + 4z} dS$, where S is part of the paraboloid $z = x^2 + y^2$ inside the cylinder $x^2 + y^2 = 2$.
8. Evaluate $\iint_S F \cdot n dS$, where
- a) $F(x, y, z) = xi - j + 2x^2k$ and S is part of the paraboloid $z = x^2 + y^2$ above the xy plane bounded by the parabolas $x = 1 - y^2$ and $x = y^2 - 1$.
- b) $F(x, y, z) = y(x^2 + y^2)^{\frac{3}{2}}i + x(x^2 + y^2)^{\frac{3}{2}}j + (z + 1)k$ and S the boundary of the solid region bounded above by the plane $z = 2x$ and below by the paraboloid $z = x^2 + y^2$.
- c) $F(x, y, z) = (3x^4 + e^{y^3})i + (x^3 + y^3)j + 12xy^2zk$ and S is the surface bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = 6x + 5$ and $z = 0$.
9. Evaluate $\iint_S (\text{curl } F) \cdot n dS$, where $F(x, y, z) = x^3yi - y^3j + z \sec(xyz)k$ and S is the portion of the sphere $\rho = \sqrt{2}$, above the plane $z = 1$, with outward normal.
10. Use Stokes' Theorem to evaluate :
- a) $\int_C F \cdot dr$, where $F(x, y, z) = (x^2 + z)i + (y^2 + x)j + (z^2 + y)k$ and C is the intersection of the sphere $\rho = 1$ and the cone $z = \sqrt{x^2 + y^2}$ with outward normal.
- b) $\oint_C F \cdot dr$, where $F(x, y, z) = 2yi + 3zj - 2xk$ and C is part of the sphere $\rho = 1$ in the first octant with outward normal.
- c) $\oint_C F \cdot dr$, where $F(x, y, z) = z^2i + xj - 3zk$ and C is the intersection of the plane $z + 2y = 6$ and the cylinder $x^2 + y^2 = 4$ with outward normal.

Assignment II (10 %), to be submitted within a week after the end of chapter 3

- Let S be the surface which is the boundary of the solid region bounded above and below by the sphere $x^2 + (y - 1)^2 + z^2 = 8$ and on the sides by the cylinder $x^2 + (y - 1)^2 = 4$ then evaluate $\iint_S F \cdot n ds$ where $F(x, y, z) = 4xi + 2yj - 5zk$ (n -directed outward)
- Find the mass of a wire that lies along the curve $r(t) = ti + (t^2 - 1)j + 3tk$, $0 \leq t \leq 1$, if the density is $\delta(t) = \left(\frac{3}{2}\right)t$. (Mass = $\int_C \delta(t) ds$)
- Evaluate $\int_C F \cdot dr$, where $F(x, y, z) = 2zi + 3xj + 4yk$ and C is the intersection of the sphere $x^2 + y^2 + z^2 = 5$ and the plane $z = -1$ in a clockwise direction.